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Medicaid Enrollees With Opioid Use Disorder Were More Likely To Receive Medication Treatment Than Commercial Enrollees

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ABSTRACT Medications for opioid use disorder (MOUD) are underused in the treatment of people with OUD; insurance-related barriers are a possible driving factor. We used Wisconsin's all-payer claims database for 2021–22 to examine differences in MOUD treatment probabilities among nonelderly adults with Medicaid insurance compared with those with commercial insurance. We found that Medicaid enrollees with OUD were 9.8 percentage points more likely to receive MOUD than commercial enrollees with OUD. Some of this gap can be explained by lower treatment rates among older enrollees and the older age distribution of commercial enrollees, as well as the fact that Medicaid enrollees are more likely to see primary care providers with more OUD experience. However, after we controlled for patient age, sex, comorbidities, and provider fixed effects, a substantial unexplained difference of 7.1 percentage points remained, potentially indicating a role for direct plan-level effects or unobserved patient characteristics. Policies aimed at improving commercial insurance coverage, training more providers, and targeting older populations may be needed to increase overall MOUD treatment rates.

Access to high-quality treatment for people with opioid use disorder (OUD) is needed to stem the overdose crisis, which resulted in more than 100,000 deaths in the US in 2023, most of which involved opioids.¹ The gold-standard treatment for OUD is one of three medications approved by the Food and Drug Administration: methadone, buprenorphine, and naltrexone.² These medications, often referred to as medications for OUD, or MOUD, halve the risk for overdose death.^{3–5} Yet their use remains persistently low;⁶ of the estimated 5.7 million Americans ages twelve and older with OUD in 2023, fewer than 20 percent received MOUD.⁷

Although the reasons for underuse of MOUD are multifactorial, health insurance coverage can shape access. Among people with substance use

disorder (SUD) who believe that they would benefit from treatment, the most commonly cited reasons for not receiving treatment include financial costs and insurance-related barriers.⁷ Prior research has found that expansions of insurance coverage and removal of insurance barriers can increase the use of MOUD.^{8–15} This article extends this research by examining whether the type of coverage a person with OUD has shapes the likelihood that they receive MOUD.

Medicaid and private insurance collectively covered more than 70 percent of people with OUD in the US as of 2019, in roughly equal proportions.¹⁶ Among people with OUD, national survey data from the period 2015–17 found that people with Medicaid were 70–90 percent more likely to receive MOUD than those with commercial insurance.^{17,18} The reasons for this large dif-

ference are unknown, but plan-, provider-, and patient-level factors could play a role. This article disentangles these potential contributors.

Having Medicaid versus commercial insurance could affect access to OUD treatment through multiple channels. First, insurance may directly affect treatment access through plan design. For example, plans might not cover all types or formulations of MOUD, or they may impose prior authorization or other limits (for example, dosage limits).^{13,19} Although cost sharing in Medicaid is generally low, commercial enrollees can face significant out-of-pocket expenses from treatment.^{20,21} Second, insurance may indirectly affect access to treatment through provider networks and reimbursement rates.^{22–25} Secret-shopper studies have found that having Medicaid insurance is negatively associated with being able to obtain a new patient appointment with a MOUD treatment provider.^{26,27} Yet providers who regularly see Medicaid patients may have more experience treating OUD and be more likely to prescribe MOUD. Finally, underlying differences in the composition of patients in these insurance market segments, such as differing levels of severity of OUD, may drive greater observed use of MOUD in Medicaid populations.

This study used Wisconsin's all-payer claims database to examine factors explaining differences in MOUD treatment rates between Medicaid and commercial insurance enrollees with OUD. Specifically, we assessed the extent to which the difference could be explained by patient demographics (age or sex) and comorbidities and by characteristics of patients' primary care providers (for example, qualification, age, and experience with OUD). Any difference that remained unexplained after we controlled for these factors may be due to direct coverage-related effects or other unobservables in our data.

Study Data And Methods

DATA Our primary data source was Wisconsin's all-payer claims database, published by the Wisconsin Health Information Organization, from the first quarter of 2021 through the first quarter of 2022. Focusing on a single state allowed us to control for state-level factors that could drive differences in national MOUD rates. The estimated prevalence of OUD among Wisconsin adults during this time frame was 2.2 percent, which is similar to the national average during 2021–22.²⁸ We also linked providers in the all-payer claims database on National Provider Identifiers to the OneKey database, a proprietary database of more than ten million health care providers and their characteristics.

The Wisconsin all-payer claims data contain information on 100 percent of the Wisconsin Medicaid population and approximately 46 percent of the commercially insured population in 2021. Lower representation in commercial data is largely because state all-payer claims databases cannot require data from self-insured plans.²⁹ Our data included an estimated 80 percent of the fully insured population and 15 percent of the self-insured population. Instead of expanding Medicaid under the Affordable Care Act, Wisconsin enacted a “partial expansion” that covers all nonelderly adults with incomes below the federal poverty level. Unlike many other all-payer claims databases,³⁰ substance use claims are not redacted in the Wisconsin all-payer claims database. The data also provide information on provider specialty and qualification (for example, doctor of medicine or doctor of osteopathic medicine [MD/DO] and nurse practitioner or physician assistant [NP/PA]).

STUDY SAMPLE Our analytic sample included people with OUD ages 18–64 who had Medicaid or commercial insurance. We defined people with OUD as those with either an OUD diagnosis in any position or a MOUD claim (procedure codes for methadone or buprenorphine administration or a pharmacy claim for buprenorphine, excluding formulations prescribed for pain) at any point during the study period. All codes are listed in online appendix table 1.³¹ We followed previous work^{32,33} by including people with MOUD claims but without an OUD diagnosis because pharmacy claims do not contain diagnosis codes. About 5 percent of our sample was identified this way. As naltrexone use was extremely rare in our sample and is also indicated for alcohol use disorder in addition to OUD, we did not include people with no OUD diagnosis who were receiving naltrexone.

We assigned each patient a primary care provider, defined as the provider with whom they had the plurality of claims during the study period among physicians, nurse practitioners, and physician assistants or associates with a specialty of family medicine, adolescent medicine, general practice, internal medicine, pediatrics, or preventive medicine. We also identified patients who did not see a primary care provider during our study period.

Our analysis was conducted at the patient-quarter level: We included patients in every quarter in which they were ages 18–64 and were continuously enrolled in either Medicaid or a commercial insurance plan with pharmacy coverage.

VARIABLES Our outcome measure was an indicator equal to 1 if a patient received methadone or buprenorphine at any point in a given quarter.

Our key independent variable of interest was an indicator for whether a patient had Medicaid or commercial insurance.

Patient-level covariates included age, sex, and ZIP code of residence. We also captured patient comorbidities that may have affected receipt of MOUD, including number of chronic condition diagnoses (using the Chronic Conditions Warehouse) and diagnoses of a nonopioid SUD, serious mental illness (schizophrenia or bipolar disorder), or chronic pain. For the subsample of people with an OUD diagnosis, we measured the severity of OUD using the modal diagnosis code appearing in their claims.

Provider-level covariates included qualification type (MD/DO or NP/PA); experience treating patients with OUD, or the proportion of all patients seen by the provider during our sample period (that is, the provider's patient panel) who had a diagnosis of OUD; and age, which may proxy for years of experience and cohort effects related to when the provider was trained. Provider age was obtained by linking the all-payer claims database on the National Provider Identifiers in OneKey; however, this information was only available for 52.6 percent of our sample.

Our models also included year-quarter fixed effects to control for secular time trends.

EMPIRICAL STRATEGY We calculated unadjusted proportions and means for all covariates and our key outcome (receipt of MOUD) for the study sample overall and stratified by whether patients had Medicaid or commercial insurance. We also computed the standardized mean difference across each variable between these two populations, calculated as the difference in the two group means divided by the square root of the average of the sample variances for each group. We then investigated rates of MOUD treatment by two covariates identified as important from our regression analyses (described next): patient age and primary care provider experience treating patients with OUD (or lack of a primary care provider). Specifically, we calculated MOUD treatment rates for different categories of age and provider experience, stratified by insurance type. We then used these age-specific and experience-specific treatment rates to compare adjusted treatment rates, where we adjusted the commercial treatment rate to mirror the age and experience composition of the Medicaid population. Finally, we used multivariable linear probability models to examine the association between Medicaid enrollment and receipt of MOUD after adjustment for a richer set of covariates.

We estimated five distinct models that sequentially added covariates: an unadjusted model with no covariates (model 1); a "patient demo-

Provider-level differences may have played a role in explaining the treatment gap.

graphics" model (model 2), which adjusted for patient age and sex; a "patient demographics and comorbidities" model (model 3), which added patient comorbidities (number of chronic conditions, indicators for diagnoses of a non-OUD substance use disorder, serious mental illness, and chronic pain); a "patient and primary care provider characteristics" model (model 4), which added characteristics of the patient's primary care provider (indicator for whether the patient saw a primary care provider, and conditional on seeing a primary care provider: provider qualification [NP/PA or MD/DO], experience with OUD, and age); and a "patient characteristics and primary care provider fixed effects" model (model 5), which replaced the primary care provider characteristics in model 4 with a fixed effect for each primary care provider.

Sensitivity analyses included adjusting for patient ZIP code fixed effects, which may proxy for unobserved patient characteristics (for example, income and race and ethnicity) and local provider supply; assigning patients to primary care providers using only their claims preceding their first observed claim with an OUD diagnosis, to alleviate concerns that patients may have selected their provider to obtain MOUD; excluding patients with no OUD diagnosis and adjusting for OUD severity; using probit models rather than linear probability models; examining methadone use and buprenorphine use separately; and using 2017–18 data.

LIMITATIONS Our study was subject to limitations. First, our results might not necessarily generalize beyond Wisconsin to the rest of the country. Second, we had incomplete coverage of the commercially insured population, particularly self-insured enrollees. Third, we did not have data on race and ethnicity. Fourth, plan information was also limited, so we were unable to measure differences in coverage, including cost sharing or whether mental health and SUD care were carved out.³³ Fifth, providers' OUD experience was measured only over our sample period and does not reflect a provider's full his-

tory. Finally, our primary analysis period occurred during the COVID-19 pandemic; we included a sensitivity analysis using 2017–18 data to examine whether findings were similar in an earlier, prepandemic period.

Study Results

Our sample contained 111,754 patient-quarters (29,719 unique individuals) for nonelderly adults with OUD and Medicaid and 11,594 patient-quarters (3,526 unique individuals) for nonelderly adults with OUD and commercial insurance (exhibit 1). A sample selection table is shown in appendix table 2.³¹ Among this sample, 60.9 percent of Medicaid enrollees with OUD

received MOUD, compared with 51.1 percent of commercial enrollees, for a difference of 9.8 percentage points (or a 19 percent higher treatment rate for Medicaid enrollees compared with commercial enrollees). A lower proportion of commercially insured enrollees with OUD received methadone compared with Medicaid enrollees (9.1 percent compared with 27.2 percent), whereas a slightly higher proportion of commercial enrollees received buprenorphine (42.6 percent compared with 35.7 percent). (In addition to percentage differences, exhibit 1 also shows the standardized mean differences for these characteristics.)

POPULATION CHARACTERISTICS Relative to the commercial population, the Medicaid popula-

EXHIBIT 1

Characteristics of study sample of nonelderly Wisconsin adults with opioid use disorder (OUD) and their primary care providers, by insurance source, 2021–22

	Medicaid (n = 111,754)	Commercial (n = 11,594)	Standardized mean difference	95% CI
MOUD treatment (%)				
Any MOUD	60.9	51.1	0.20	0.18, 0.22
Buprenorphine	35.7	42.6	−0.14	−0.16, −0.12
Methadone	27.2	9.1	0.48	0.47, 0.50
Patient characteristics				
Age, years (%)				
18–29	17.1	16.2	0.02	0.00, 0.04
30–39	41.0	30.7	0.22	0.20, 0.23
40–49	23.1	24.3	−0.03	−0.05, −0.01
50–64	18.8	28.8	−0.24	−0.25, −0.22
Sex (%)			0.27	0.25, 0.30
Female	56.4	43.0		
Male	43.6	57.0		
No. of chronic conditions per person	0.43	0.42	0.01	−0.01, 0.03
Non-OUD SUD (%)	49.3	31.7	0.36	0.34, 0.38
Serious mental illness ^a (%)	17.8	8.1	0.29	0.27, 0.31
Chronic pain (%)	14.2	17.2	−0.08	−0.10, −0.06
Saw any primary care provider (%)	90.7	89.0	0.06	0.04, 0.08
Primary care provider characteristics (%)				
Qualification				
NP/PA	41.2	25.5	0.34	0.32, 0.36
MD/DO	58.8	74.5	−0.34	−0.36, −0.32
Patient panel share with OUD				
<2.5%	22.4	50.3	−0.61	−0.63, −0.59
2.5–6.5%	25.6	19.3	0.15	0.13, 0.17
6.6–42.1%	26.3	18.6	0.18	0.17, 0.20
>42.1%	25.8	11.9	0.36	0.34, 0.38
Provider age, years				
Younger than 40	9.8	11.5	−0.05	−0.07, −0.03
40–59	30.6	43.4	−0.27	−0.29, −0.25
60 or older	13.2	14.5	−0.04	−0.06, −0.02
Missing	46.4	30.6	0.33	0.31, 0.35

SOURCE Authors' analysis of data from the Wisconsin all-payer claims database, first quarter 2021 through first quarter 2022. **NOTES** Sample sizes are patient-quarter observations. Patients must have been enrolled continuously in either Medicaid or commercial insurance for the entire quarter and had OUD or received medications for OUD (MOUD) at some point in the sample. The table shows the main outcome, MOUD treatment, a binary variable indicating receipt of buprenorphine or methadone in that quarter, as well as patients' and primary care providers' characteristics. Standardized mean differences were calculated as described in the text. SUD is substance use disorder. NP/PA is nurse practitioner or physician assistant. MD/DO is doctor of medicine or doctor of osteopathic medicine. ^aSchizophrenia or bipolar disorder.

tion was significantly younger (58.1 percent younger than age forty in the Medicaid group compared with 46.9 percent in the commercial group), more likely to be female (56.4 percent compared with 43.0 percent), more likely to have a non-OD SUD (49.3 percent versus 31.7 percent), and more likely to have serious mental illness (17.8 percent versus 8.1 percent). Similar proportions of Medicaid and commercial enrollees saw a primary care provider during our study period (90.7 percent versus 89.0 percent). Among them, Medicaid enrollees were more likely than commercially insured enrollees to have a primary care provider who was an NP/PA (41.2 percent versus 25.5 percent), had more experience with OUD, and was younger, on average.

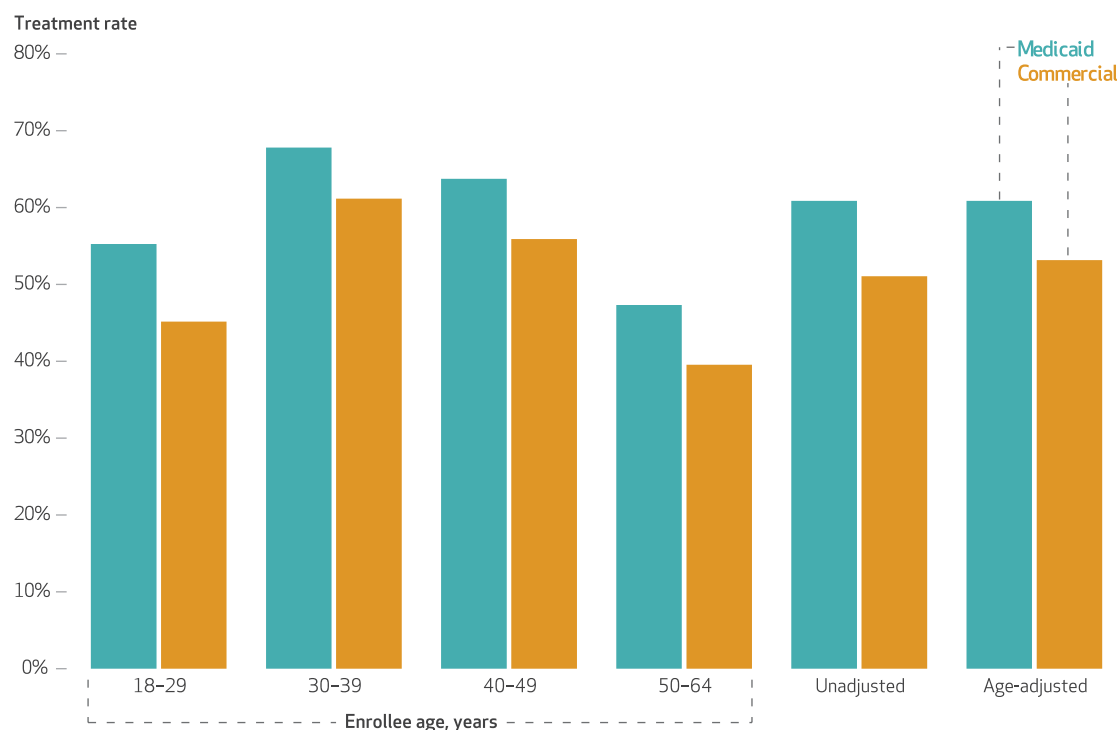
MOUD TREATMENT RATES Among both Medicaid and commercial enrollees with OUD, MOUD treatment rates were highest for people ages 30–39 and 40–49 and were lower for the youngest and oldest age groups (ages 18–29 and 50–64; exhibit 2). However, within age groups, MOUD treatment rates remained higher for Medicaid enrollees than for commercial enrollees. If the age distribution of commercial enrollees re-

flected the age distribution of Medicaid enrollees but treatment rates remained the same within age-by-insurance market groups (the “age-adjusted” bars in exhibit 2), the difference in MOUD treatment rates would be reduced from 9.8 percentage points to 7.7 percentage points, reflecting the fact that the older group was disproportionately represented among commercial enrollees and was the least likely to receive MOUD treatment.

The difference in MOUD treatment rates between Medicaid and commercial enrollees was greatest among people whose primary care providers had more experience treating patients with OUD (exhibit 3). MOUD treatment rates generally increased with providers’ experience with OUD, but a significant gap between Medicaid and commercial enrollees existed within provider-experience groups. Among Medicaid enrollees, the MOUD treatment rate increased from 51 percent for those seeing primary care providers with the lowest proportions of OUD in their patient panels to 85 percent for those seeing primary care providers with the most OUD experience. In contrast, commercial enrollees’ treatment rates increased from 44 percent to

EXHIBIT 2

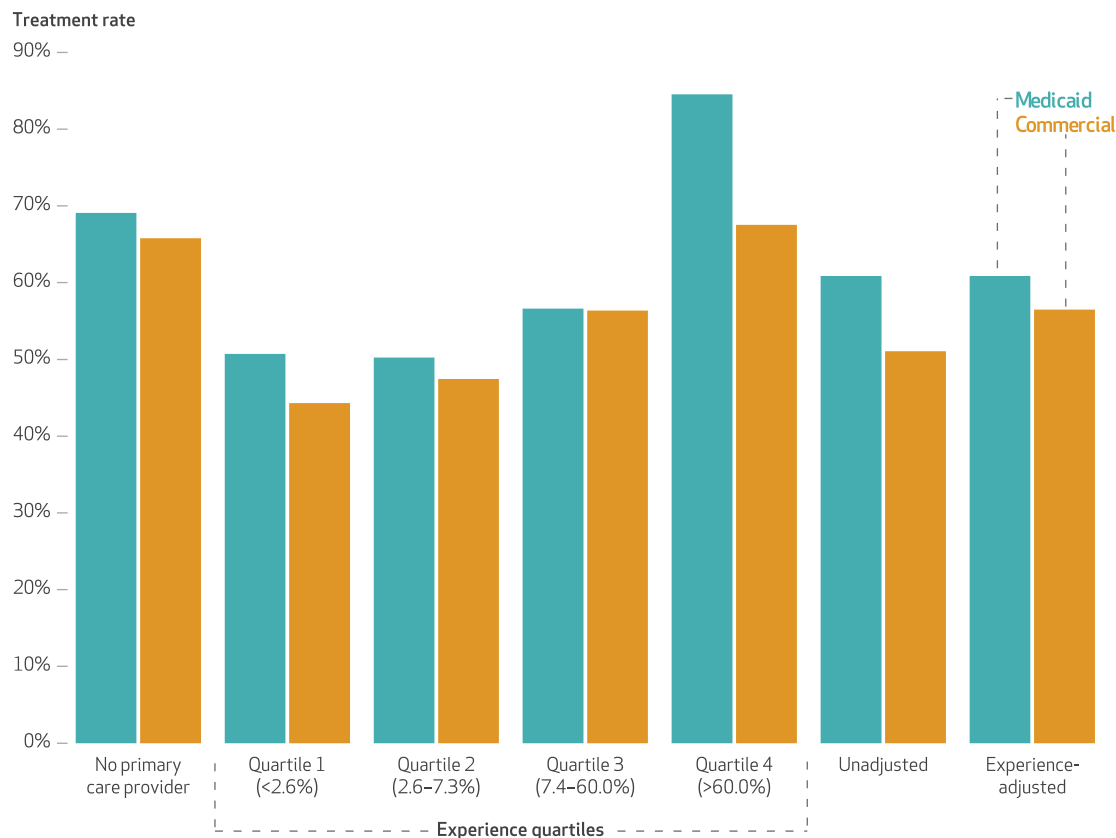
Medications for opioid use disorder (MOUD) treatment rate among nonelderly Wisconsin adults with OUD, by enrollee age and insurance source, 2021–22



SOURCE Authors’ analysis of data from the Wisconsin all-payer claims database, first quarter 2021 through first quarter 2022. **NOTE** The age-adjusted MOUD treatment rate for commercial insurance patients is the predicted treatment rate if the age distribution of commercial enrollees reflected the age distribution of Medicaid patients.

EXHIBIT 3

Medications for opioid use disorder (MOUD) treatment rate among nonelderly Wisconsin adults with OUD, by quartiles of OUD prevalence in providers' patient panels and insurance source, 2021–22



SOURCE Authors' analysis of data from the Wisconsin all-payer claims database, first quarter 2021 through first quarter 2022. **NOTE** The experience-adjusted MOUD treatment rate for commercial insurance patients is the predicted treatment rate if the provider experience distribution of commercial enrollees reflected the distribution of provider experience for Medicaid patients.

68 percent across primary care provider experience categories. Treatment rates for people without primary care providers were 69 percent and 66 percent for Medicaid and commercial enrollees, respectively. Because Medicaid enrollees were also more likely to see more experienced providers, adjusting for having a primary care provider and primary care provider experience reduced the gap between Medicaid and commercial enrollees from 9.8 percentage points to 4.4 percentage points.

The MOUD treatment gap was reduced after adjustment for patient and primary care provider characteristics (exhibit 4). Controlling for patient demographics (age and sex, model 2) cut the difference to 8.1 percentage points (95% confidence interval: 7.2, 9.0), down from 9.8 percentage points (95% CI, 8.9, 10.7). However, adding patient comorbidities (model 3) increased the gap. This increase reflects the fact that people with serious mental illness and other co-occurring SUDs are disproportionately represented in Medicaid and are also less likely to

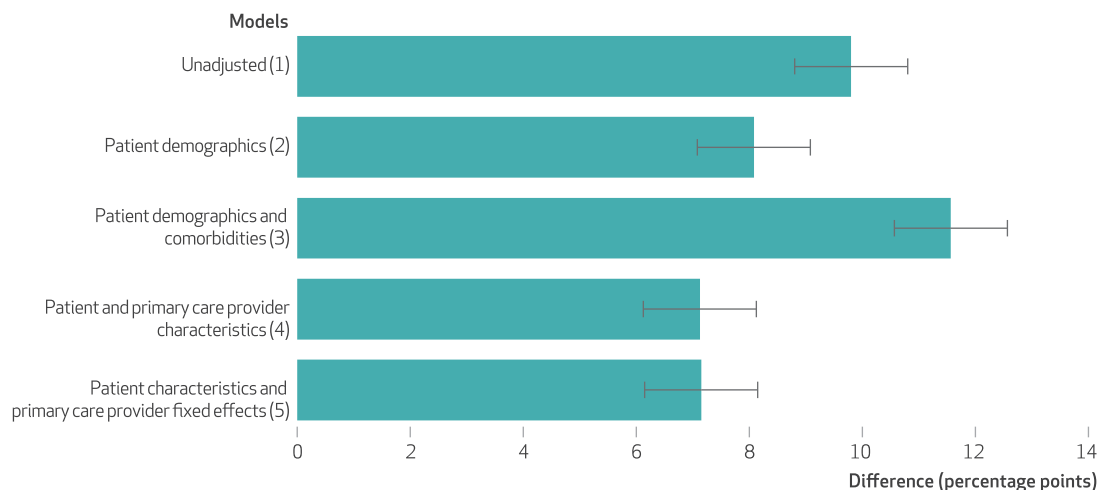
receive MOUD across both insurance markets (appendix table 3).³¹

Adding primary care provider characteristics, including having a primary care provider and primary care provider age, qualification, and experience with OUD, to the model (model 4) reduced the gap to 7.1 percentage points (95% CI: 6.3, 8.0). Adding primary care provider fixed effects in lieu of characteristics (model 5) had a similar effect, reducing the gap to 7.1 percentage points (95% CI: 5.3, 9.0). Thus, the most saturated model was able to explain about 28 percent of the unadjusted gap in treatment rates between Medicaid and commercially insured enrollees, meaning that a sizable gap remained unexplained after patient and provider characteristics were controlled for.

SENSITIVITY ANALYSES Findings were very similar when we included patient ZIP code fixed effects in all adjusted models (appendix table 4)³¹ and when we assigned patients' primary care providers using only their pre-OUD claims to avoid potential endogenous primary care provid-

EXHIBIT 4

Difference in medications for opioid use disorder (MOUD) treatment rates among nonelderly Wisconsin adults with OUD for Medicaid patients compared with commercially insured patients, estimated from models with different patient- and provider-level controls, 2021–22



SOURCE Authors' analysis of data from the Wisconsin all-payer claims database, first quarter 2021 through first quarter 2022. **NOTES** Bars represent estimated coefficient estimates from five ordinary least squares regression models with MOUD treatment as the dependent variable, having Medicaid insurance as an independent variable of interest (reference group: commercial insurance), and sequentially added controls. Model 1 (unadjusted) contains no covariates; models 2–5 are described in more detail in the text. The dependent variable, MOUD treatment, is expressed in percentage points. Whiskers indicate 95% confidence intervals. Full regression results are in appendix table 3 (see note 31 in text).

er choice (appendix table 5).³¹ Excluding patients with no OUD diagnosis and controlling for OUD severity resulted in a larger estimated unadjusted gap (13.1 percentage points), shrinking to 7.1 percentage points in the most saturated models (appendix table 6).³¹ Patterns were similar when we used probit models (appendix table 7).³¹ The unadjusted findings of increasing methadone use and a smaller decrease in buprenorphine use among Medicaid enrollees were robust to adjusted models (appendix tables 8 and 9).³¹ Analysis of data from the 2017–18 period found lower overall treatment rates compared with the 2021–22 period and an even larger gap between Medicaid and commercial insurance enrollees (appendix table 10).³¹ In this period, younger age and greater primary care provider experience with OUD similarly predicted greater likelihood of receiving MOUD.

Discussion

Consistent with national survey data,^{17,18} this study documented a stark disparity in the proportion of Medicaid and commercial insurance enrollees with OUD receiving MOUD in Wisconsin during 2021–22. Sixty-one percent of Medicaid enrollees with OUD received MOUD in any given quarter, compared with 51 percent of commercial enrollees with OUD. This disparity may be surprising, given that Medicaid

serves a low-income population that may face greater barriers to care.^{34,35} Similar to other claims-based analyses,^{32,33} these treatment rates are likely higher than estimates from survey data because of the need for a medical provider to submit a claim with an OUD diagnosis for a patient to be counted in the denominator.

This treatment gap is partially explained by patients' age: Patients ages 50–64 enrolled in both insurance types were the least likely to receive treatment and also represented a greater share of commercial enrollees. However, a significant gap remained even after we controlled for patients' age and characteristics more broadly. Adjusting for sex and patient comorbidities increased the estimated gap as a result of lower overall rates of MOUD among women and people with co-occurring SUDs or serious mental illness—populations that are disproportionately represented among Medicaid enrollees.

Provider-level differences may also have played a role in explaining the treatment gap. Medicaid enrollees were more likely to see providers who had greater experience treating OUD, measured via the composition of their patient panels. However, conditional on seeing these more experienced providers, Medicaid and commercial enrollees still received treatment at different rates.

After these patient- and provider-level differences were accounted for, 72 percent of the gap

The finding of low treatment rates among older populations should motivate an increased focus on these populations.

remained unexplained, suggesting that direct insurance factors or unobserved patient characteristics may explain lower rates of treatment among commercial enrollees relative to Medicaid enrollees. Unfortunately, we could not observe cost sharing, prior authorization requirements, or coverage decisions by commercial plans in our data, although prior work has found evidence of such restrictions.³⁶

We also found high rates of methadone use among the Medicaid population compared with very low rates among commercial insurance enrollees. Commercial plans might not cover methadone,^{20,37} which may have driven this difference. Importantly, we could observe insurance-paid treatment only; commercial enrollees may be paying fully out of pocket for methadone. It is also impossible to know whether the slightly lower rates of buprenorphine use in Medicaid populations were due to substitution effects from greater methadone use or to independent factors that reduced the use of buprenorphine in Medicaid. This is an important question for future research, given that some patients may have difficulty accessing methadone because of the need to receive medication in opioid treatment programs.

Our results have multiple implications for policy makers. First, the treatment disparity between Medicaid and commercial insurance enrollees suggests that although Medicaid expansion and other Medicaid policies may have increased access for many,^{8–11} commercial enrollees might not have experienced the same

gains.^{33,38} Although coverage has improved in both Medicaid and commercial plans in the past decade,^{36,39} state and federal policy makers could consider requiring greater coverage of MOUD among commercial plans.⁴⁰

Second, our finding of significant provider-level effects on MOUD treatment rates indicates that efforts to train more providers and expand the set of providers who are comfortable referring or offering MOUD treatment may meaningfully improve treatment rates. Recent policies such as the elimination of the X waiver in the Consolidated Appropriations Act, 2023, could be an important step in the right direction.⁴¹

Third, our finding of high rates of methadone use among Medicaid enrollees may indicate that offering a choice of medications may increase overall treatment rates. Coverage of methadone in Medicaid programs has been universal since the passage of the Substance Use Disorder Prevention that Promotes Opioid Recovery and Treatment for Patients and Communities (SUPPORT) Act of 2018; similar efforts for the commercial insurance population may be warranted.

Finally, the finding of low treatment rates among older populations should motivate an increased focus on these populations, since people older than age fifty-five were the only age groups to experience increases in overdose death rates from 2022 to 2023.⁴²

Conclusion

Nonelderly Medicaid enrollees with OUD in Wisconsin were 19 percent more likely to receive MOUD than their commercially insured counterparts during the period 2021–22. This difference can be explained in part by the older age composition of commercially insured enrollees and by the fact that these enrollees were less likely to see primary care providers with substantial exposure to OUD. However, a substantial unexplained difference remained after adjustment for observable patient and provider characteristics. Policy makers should consider policies that address gaps in commercial insurance coverage of MOUD, particularly of methadone; that educate and train more providers in best practices for treating OUD; and that improve access to care for older adults. ■

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